

Table 1: Shared Training Hyperparameters

Parameter	Value
Base model	Meta-Llama-3.1-8B-Instruct
Fine-tuning method	QLoRA
LoRA rank	8
LoRA alpha	16
LoRA dropout	0.05
Quantization	4-bit bitsandbytes (bnb)
Training stage	Supervised fine-tuning (SFT)
Optimizer	paged_adamw_32bit
Learning-rate scheduler	cosine
Number of epochs	3
Gradient accumulation steps	8
Gradient checkpointing	enabled
Template	llama3
Precision	bf16